

IPO Patent Form 2

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FORM 2 [FILING PREPARATION DRAFT]

PROVISIONAL / COMPLETE SPECIFICATION (Section 10; Rule 13)

1. Title of Invention:

AI-Powered Adaptive Traffic Signal Optimization Using Real-Time Traffic Prediction
2. Specification Type:

COMPLETE
3. Preamble to Description:

The following specification particularly describes the invention and the manner in which it is to be performed.

4. DESCRIPTION & CLAIMS:

Abstract Description:

Innovation Abstract

The proposed invention is an AI-powered adaptive traffic signal optimization system designed to dynamically control traffic signals based on real-time and predicted traffic conditions. Unlike conventional traffic signal systems that operate using fixed timing schedules or predefined traffic rules, the proposed system continuously analyzes traffic conditions and automatically determines optimized signal timings.

The system collects traffic information from cameras, sensors, connected vehicles, and historical traffic datasets. AI-based computer vision and machine learning techniques are used to estimate vehicle density, queue length, traffic flow, waiting time, and congestion levels at individual road segments and intersections.

The system further predicts short-term traffic conditions using historical and real-time traffic patterns. Based on the predicted traffic state, an optimization mechanism dynamically adjusts signal phases, green-light duration, red-light duration, and phase priorities.

The system can also coordinate multiple interconnected traffic signals rather than optimizing each intersection independently. This enables traffic flow to be considered across a larger road network and reduces the propagation of congestion between neighboring intersections.

The proposed invention therefore provides an adaptive traffic management mechanism capable of continuously sensing, predicting, optimizing, and modifying traffic signal behavior according to changing traffic conditions.

Problem Statement:

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Conventional traffic signal systems generally operate using fixed timing schedules or predetermined traffic rules. These systems cannot effectively respond to sudden changes in traffic volume, unexpected congestion, accidents, road blockages, weather conditions, or variations in traffic demand.

As a result, vehicles may experience unnecessary waiting time even when one direction has significantly higher traffic demand than another. Independent operation of traffic signals can also create congestion between adjacent intersections and produce inefficient traffic flow across an entire road network.

Existing adaptive systems may respond to current traffic conditions but may not sufficiently incorporate short-term traffic prediction and coordinated optimization across multiple intersections.

Proposed Solution:

Proposed Technical Solution

The proposed system operates through the following technical process:

1. **Traffic Data Collection:**

Traffic information is collected from roadside cameras, traffic sensors, connected devices, and historical traffic datasets.

2. **Traffic Detection:**

Computer vision and sensor-processing techniques identify vehicles and estimate traffic parameters such as vehicle count, queue length, vehicle speed, lane occupancy, and traffic density.

3. **Real-Time Traffic State Generation:**

The detected information is converted into a structured representation of the current traffic condition at each monitored intersection.

4. **Traffic Prediction:**

A machine learning prediction module analyzes historical and real-time traffic patterns to estimate short-term future traffic conditions.

5. **Congestion Assessment:**

The system calculates congestion indicators based on traffic density, queue length, waiting time, vehicle flow, and predicted demand.

6. **Signal Optimization:**

An optimization engine determines suitable signal phase sequences and dynamically calculates green-light duration, red-light duration, and phase priorities.

7. **Multi-Intersection Coordination:**

Traffic conditions from neighboring intersections are considered together to coordinate signal timings and reduce congestion propagation across connected roads.

8. **Dynamic Signal Control:**

The optimized signal configuration is transmitted to the traffic signal controller.

9. **Continuous Monitoring:**

The system continuously receives new traffic information and evaluates the effect of the current signal configuration.

10. **Adaptive Reoptimization:**

When traffic conditions change, the system recalculates the traffic prediction and signal configuration, creating a continuous feedback loop between traffic sensing, prediction, optimization, and signal control.

Novel Features:

Novel Features Mapped

Real-Time Multi-Parameter Traffic State Generation

The system simultaneously evaluates vehicle density, queue length, lane occupancy, vehicle speed, traffic flow, and waiting time to generate a comprehensive real-time traffic state.

Short-Term Traffic Prediction for Signal Control

Instead of responding only to current traffic conditions, the system predicts near-future traffic demand using historical and real-time traffic data and uses the prediction during signal optimization.

Predictive Congestion Prevention

The system identifies traffic conditions that are likely to produce congestion and proactively modifies signal timings before excessive queue formation occurs.

Dynamic Signal Phase Optimization

Green duration, red duration, phase sequence, and phase priority are dynamically calculated according to changing traffic demand rather than relying on fixed timing configurations.

Multi-Intersection Coordinated Optimization

The system considers traffic conditions at neighboring intersections and coordinates signal

Claims Count:

3

Claims Scope:

1. An automated intelligent ai-powered adaptive traffic signal optimization using real-time traffic prediction system, comprising: (a) a distributed sensor module configured to capture real-time operational environmental parameters;
(b) an edge computing processor coupled to the sensor module to execute adaptive analytics;
(c) a communication interface transmitting diagnostic status telemetries to a centralized registry.
2. The system of claim 1, wherein: the edge computing processor further executes real-time machine-learning anomaly detection based on statistical deviation thresholds.
3. The system of claim 1, wherein: the communication interface encrypts telemetries utilizing asymmetric cryptographic handshakes before dispatch.